Lecture One

1 Real Functions

1.1 Introduction

The relation between two sets X and Y is a set of ordered pairs each of the form (x, y) where $x \in X$ and $y \in Y$. A **function** from X to Y is a relation between X and Y that has the property that any two ordered pairs with the same $x-$value also have the same $y-$value. Then we called x **the independent variable** and y **the dependent variable**.

Definition 1.1.1 ***Real-Valued function of a Real Variable:***
Let X and Y be sets of real numbers. A ***real-valued function*** *f of a real variable x from X to Y is a correspondence that assigns to each number x in X exactly one number in Y.*

$$X \equiv The\ domain\ of f$$
$$Y \equiv The\ Co-domain$$

Note: The number y is the image of x under f and is denoted by $f(x)$ which is called the value of f at x. The **range** of f is a subset of Y and consists of all images of numbers in X.

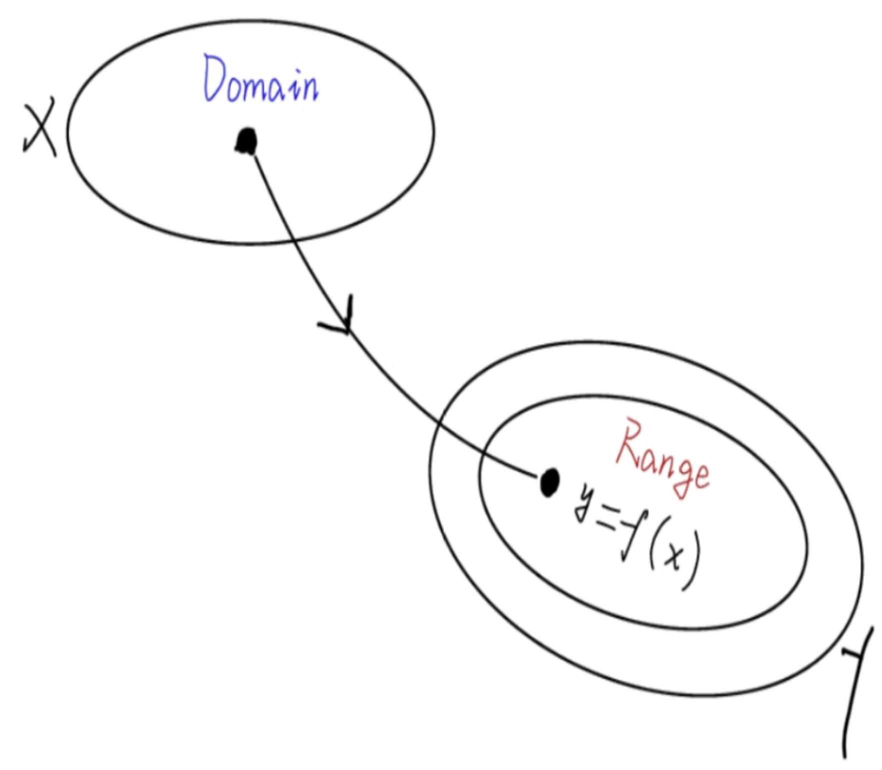


Figure 1: A real-valued function f of a real variable

For example: The area A of a circle is a function of the circle's radius r, and written as:

$$A = \pi r^2$$

Where

$$A \equiv dependent\ variable$$
$$r \equiv independent\ variable$$

Example 1.1.1 *Given $f(x) = x^2 + 3$, evaluate the following:*

1. $f(2)$
2. $f(-1)$
3. $f(3a)$
4. $f(1-a)$
5. $\dfrac{f(x+\Delta x) - f(x)}{\Delta x} \qquad , \Delta x \neq 0$

Sol

1. $f(2) = (2)^2 + 3 = 4 + 3 = 7$
2. $f(-1) = (-1)^2 + 3 = 1 + 3 = 4$
3. $f(3a) = (3a)^2 + 3 = 9a^2 + 3.$
4. $f(1-a) = (1-a)^2 + 3 = a^2 - 2a + 4.$
5. $\dfrac{f(x+\Delta x) - f(x)}{\Delta x} = \dfrac{(x+\Delta x)^2 + 3 - (x^2+3)}{\Delta x} = 2x + \Delta x$

1.2 Types of Functions

1. **Polynomial Functions:** is a function of the form $y = f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n$ where $a_0, a_1, \ldots, a_n$ are constants. For example $y = ax^3 + bx^2 + cx + d$ is a polynomial function from the third degree.

2. **Algebraic Functions:** is a function of the form $P_0(x)y^n + P_1(x)y^{n-1} + \cdots + P_n(x) = 0$ where $P_0(x), P_1(x), \ldots, P_n(x)$ are polynomials and all powers of x, y are in $\mathbf{Z}^+$.
For example $x^2y^3 + xy^2 - 2y + x + 7 = 0$

3. **Rational Functions:** is of the form:
$$f(x) = \frac{P(x)}{R(x)}$$
is called rational function, where $P(x), R(x)$ are polynomials. For example
$$f(x) = \frac{2x^2 + 3x + 1}{1 - 4x}$$

4. **Trigonometric Functions:**
   (i) $y = \sin x$
   (ii) $y = \cos x$
   (iii) $y = \tan x = \dfrac{\sin x}{\cos x}$
   (iv) $y = \cot x = \dfrac{\cos x}{\sin x}$
   (v) $y = \sec x = \dfrac{1}{\cos x}$
   (vi) $y = \csc x = \dfrac{1}{\sin x}$

5. **Inverse Trigonometric Functions:**
   (a) $\sin y = x \iff y = \sin^{-1} x$
   (b) $\cos y = x \iff y = \cos^{-1} x$
   (c) $\tan y = x \iff y = \tan^{-1} x$
   (d) $\cot y = x \iff y = \cot^{-1} x$
   (e) $\sec y = x \iff y = \sec^{-1} x$
   (f) $\csc y = x \iff y = \csc^{-1} x$

6. **Exponential Functions:**
   (i) $y = f(x) = a^x$ ($a \equiv$ constant)
   (ii) $y = f(x) = e^x$ ($e = 2.71828$)

   where $a \equiv$ the base and $x \equiv$ exponent

7. **Logarithmic Functions:**
   (i) $y = \log_a x \qquad ,a > 1$
   (ii) $y = \log_{10} x = \log x$
   (iii) $y = \log_e x = \ln x$

   Every exponential form can be transformed to logarithmic form as:
$$y = \log_a x \iff x = a^y$$
$$y = \ln x \iff x = e^y$$

8. **Hyperbolic Functions:**
   (a) $y = \sinh x$
   (b) $y = \cosh x$

(c) $y = \tanh x = \dfrac{\sinh x}{\cosh x}$

(d) $y = \coth x = \dfrac{\cosh x}{\sinh x}$

(e) $y = sech\ x = \dfrac{1}{\cosh x}$

(f) $y = csch\ x = \dfrac{1}{\sinh x}$

9. **Inverse Hyperbolic Functions:**

(a) $\sinh y = x \iff y = \sinh^{-1} x$

(b) $\cosh y = x \iff y = \cosh^{-1} x$

(c) $\tanh y = x \iff y = \tanh^{-1} x$

(d) $\coth y = x \iff y = \coth^{-1} x$

(e) $sech\ y = x \iff y = sech^{-1} x$

(f) $csch\ y = x \iff y = csch^{-1} x$

1.3 The Domain and Range of a Function:

The domain of definition for the functions:

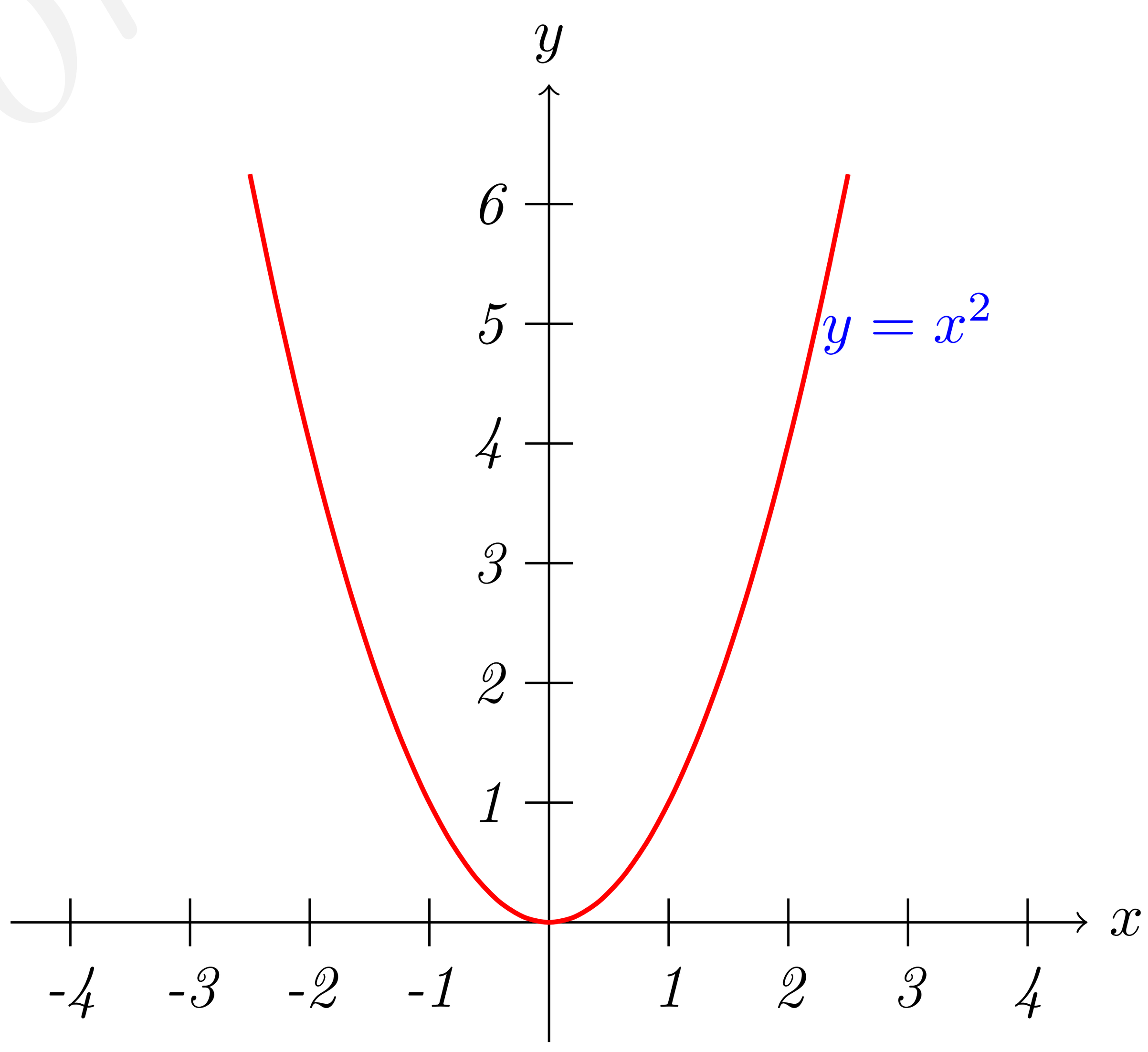
1. Polynomial functions
2. $y = \sin x$ and $y = \cos x$
3. The rational function $f(x) = \dfrac{P(x)}{R(x)}$ where $R(x)$ cannot be factorised on $\mathbb{R}$ and $P(x)$ is polynomial

is **the set of real numbers** $\mathbb{R}$ and we write it as

$$D = \mathbb{R} \text{ or } D = (-\infty, \infty) \text{ or } D =]-\infty, \infty[$$

Example 1.3.2 *Find the domain and the range of the function* $y = x^2$

Sol

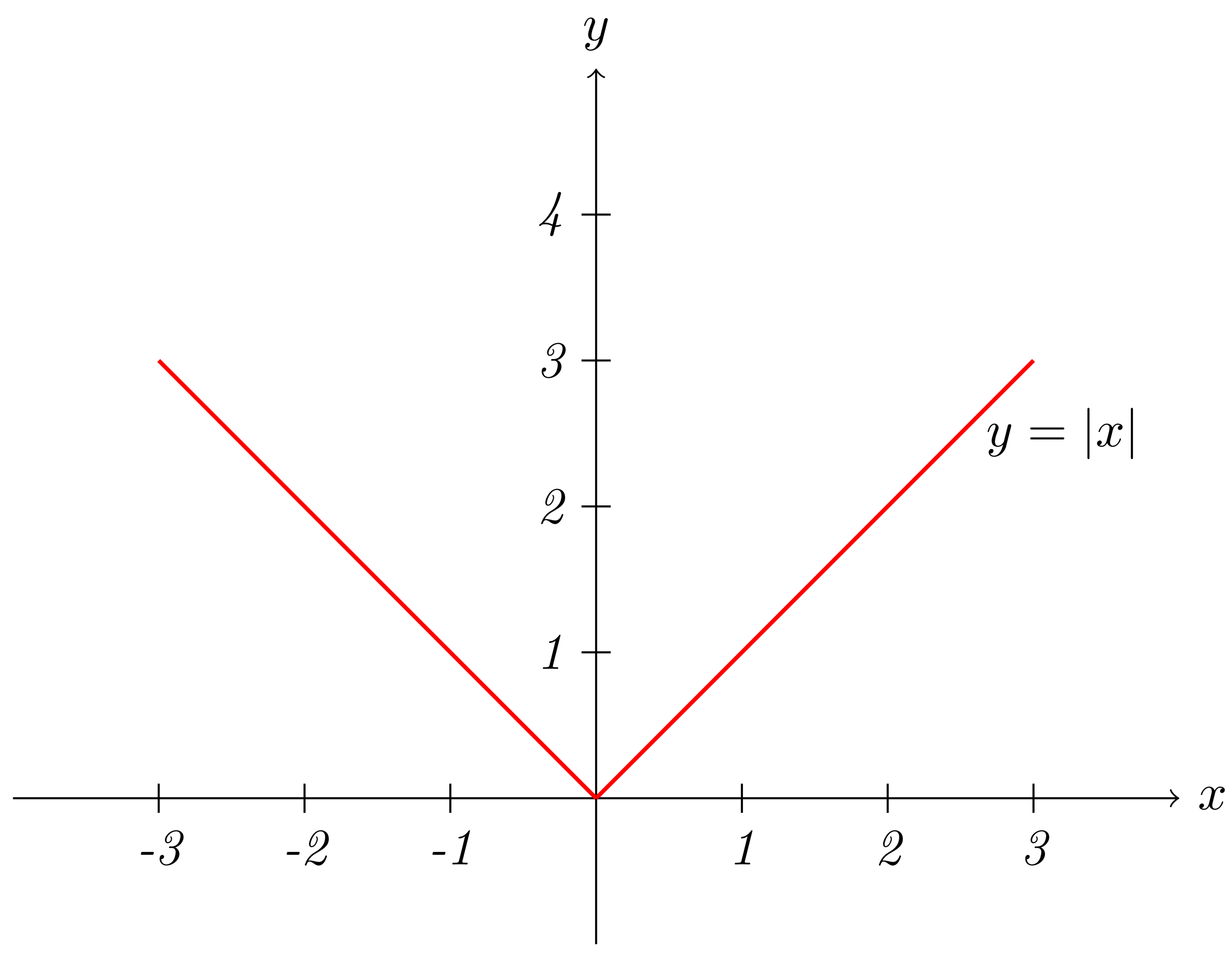


Clearly $y = x^2$ *is a polynomial function and* $\forall x = a \in \mathbb{R}, f(a) \in \mathbb{R}$. *i.e.* $D = \mathbb{R} = (-\infty, \infty)$.
The range of $y = x^2$ *is all of the positive numbers such that* $0 \leq x < \infty \equiv [0, \infty)$.

$$\therefore D = \mathbb{R} \text{ and } \mathcal{R} = [0, \infty)$$

Example 1.3.3 *Find the domain and the range of the function* $y = |x|$

Sol

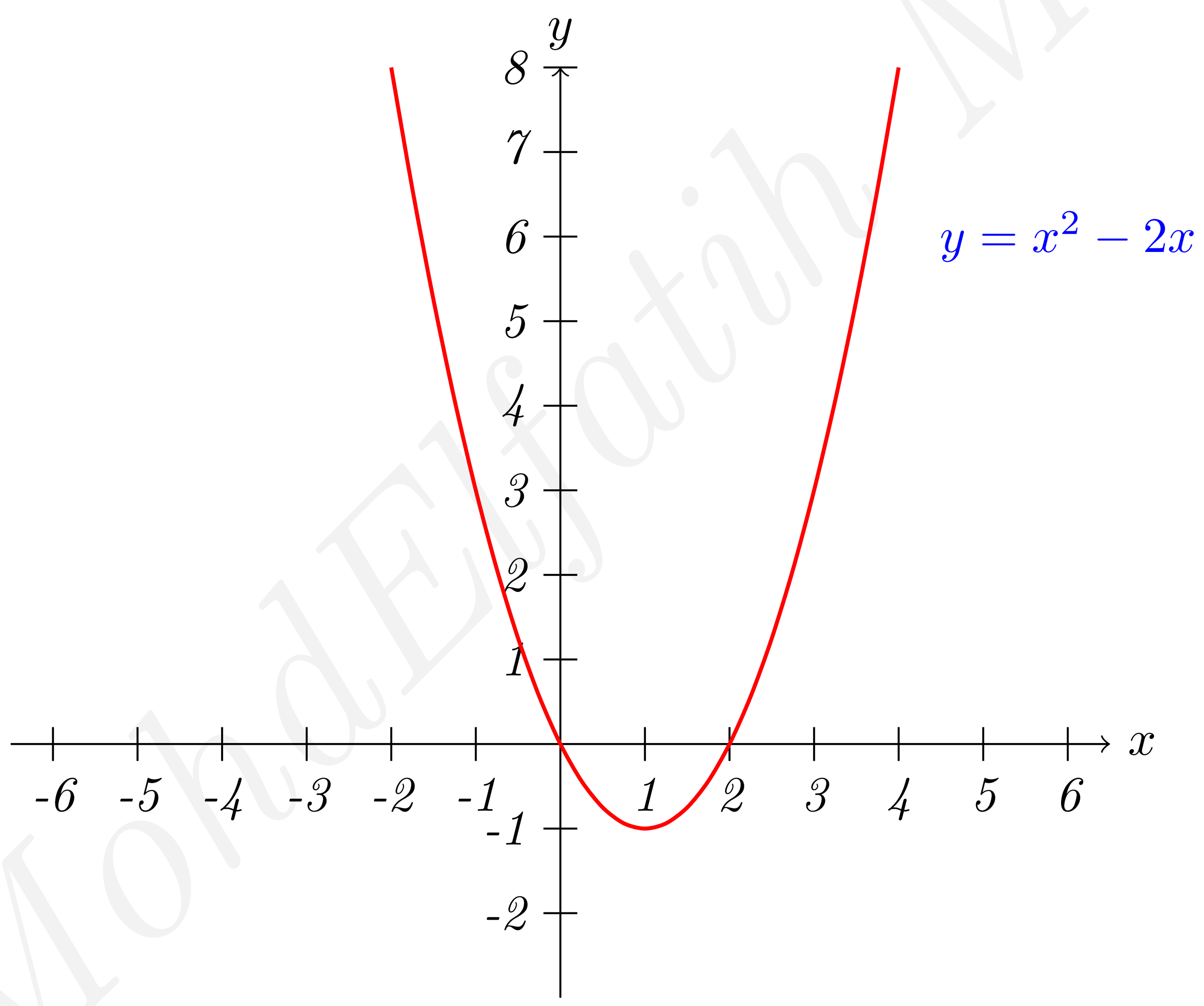


$$\forall x \in \mathbb{R}, \exists\ y \in \mathbb{R}\ s.t\ y \in [0, \infty)$$

Therefore $D = \mathbb{R}$ *and* $\mathcal{R} = [0, \infty)$

Example 1.3.4 *Find the domain and the range of the function* $y = x^2 - 2x$

Sol



Clearly, the given function is polynomial, therefore the domain is $\mathbb{R}$.

$$D = (-\infty, \infty)$$

To find the range:

$$y = x^2 - 2x + 1 - 1 = (x - 1)^2 - 1$$

The minimum value of $y = -1$ *when* $(x - 1)^2 = zero$ *at* $(x = 1)$, *thus* $y \geq -1$.
$\Rightarrow D = \mathbb{R}\ ,\ \mathcal{R} = [-1, \infty)$

Example 1.3.5 *Find the domain and the range of the function* $y = \dfrac{2}{x - 3}$

Sol

The given function is rational $\Rightarrow$ $x - 3 = 0$ $\Rightarrow$ $x = 3$

Thus $D = \mathbb{R} - \{3\}$ « $[\ \mathbb{R} - \{\text{zeros of denominator}\}\ ]$ »

Now: To find the range :

$$y = \frac{2}{x-3} \Rightarrow x - 3 = \frac{2}{y} \Rightarrow \boxed{x = \frac{2}{y} + 3} \Rightarrow y \in \mathbb{R} - \{0\} \quad \#$$

Example 1.3.6 *Find the domain and the range of the function* $\sqrt{2x^2+x}$

Sol

$y = \sqrt{x(2x+1)}$; $y \in \mathbb{R}$ only in case of $x(2x+1) \geq 0$

① x and $(2x+1)$ are +ve :

$\Rightarrow x \geq 0 \ \wedge \ 2x+1 \geq 0$

$\Rightarrow x \geq 0 \ \wedge \ x \geq -\frac{1}{2}$

$\Rightarrow [-\frac{1}{2}, \infty) \cap [0, \infty) = [0, \infty)$

② $x \leq 0 \ \wedge (2x+1) \leq 0$ [both -ve]

$\Rightarrow x \leq 0 \ \wedge \ x \leq -\frac{1}{2}$

$\Rightarrow (-\infty, 0] \cap (-\infty, -\frac{1}{2}] = (-\infty, -\frac{1}{2}]$

therefore, we conclude that

$x \in (-\infty, -\frac{1}{2}] \cup [0, \infty)$

* Range «exercise»

1.4 Combinations of Functions

Two functions can be combined in various ways to create new functions.

Example 1.4.7 *Let $f(x) = 2x - 3$ and $g(x) = x^2 + 1$, we can form the functions*

1. ***Sum***

$$(f+g)(x) = f(x) + g(x) = (2x-3) + (x^2+1) = x^2 + 2x - 2$$

2. ***Difference***

$$(f-g)(x) = f(x) - g(x) = (2x-3) - (x^2+1) = 2x - x^2 - 4$$

3. ***Product***

$$(fg)(x) = f(x)g(x) = (2x-3)(x^2+1)$$

4. ***quotient***

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{2x-3}{x^2+1}$$

Also we can combine two functions in yet another way, called **composition**.

Definition 1.4.2 ***Composition of Functions***
Let f and g be functions of x, the function given by $(f \circ g)(x) = f(g(x))$ is called the composite of f with g.

Note:

1. Domain of $(f \circ g)$ is the set of all x in the domain of g such that $g(x)$ is in the domain of f.

2. $(f \circ g)$ may not be equal to $(g \circ f)$.

Example 1.4.8 *Given $f(x) = 2x - 1$, $g(x) = \cos x$, find each of the following:*

(a) $f \circ g$

(b) $g \circ f$

(c) $(f \circ g)\left(\frac{\pi}{3}\right)$

Sol

ⓐ $f \circ g = f[g(x)] = 2(\cos x) - 1 = \underline{2\cos x - 1}$

ⓑ $g \circ f = g[f(x)] = \underline{\cos(2x-1)}$

ⓒ $(f \circ g)(\frac{\pi}{3}) = f[g(\pi/3)] = f[1/2] = 2(1/2) - 1 = 1 - 1 = \underline{0}$

1.5 Even and Odd function

1. **Even Function**: If $f(x)$ satisfied the condition $f(-x) = f(x)$ we called it "even function". For example $y = f(x) = x^2$

$$f(-x) = (-x)^2 = (-1)^2x^2 = x^2 = f(x) \Rightarrow f(-x) = f(x)$$

Note: the functions $f(x) = \cos x$ and $f(x) = secx$ are even.

2. **Odd Functions**: If $f(x)$ satisfied the condition $f(-x) = -f(x)$ we called it "odd function". For example $y = f(x) = x^3$

$$f(-x) = (-x)^3](-1)^3x^3 = -x^3 = -f(x) \Rightarrow f(-x) = -f(x)$$

Note: the functions $y = \sin x$, $y = \tan x$, $y = \cot x$, $y = \csc x$ are odd.

Example 1.5.9 *Determine whether each function is even, or odd or neither.*

(i) $f(x) = x^3 - x$

(ii) $g(x) = 1 + \cos x$

(iii) $h(x) = \dfrac{\tan x}{1 + \tan x}$

Sol

(i) $f(x) = x^3 - x$

$\Rightarrow f(-x) = (-x)^3 - (-x)$

$= -x^3 + x = -(x^3 - x) = -f(x)$

$\therefore f(-x) = -f(x) \Rightarrow$ (odd)

(ii) $g(-x) = 1 + \cos(-x)$

$= 1 + \cos x = g(x)$

$\Rightarrow g(-x) = g(x)$ (even) #

(iii) $h(-x) = \dfrac{\tan(-x)}{1 + \tan(-x)}$

$= \dfrac{-\tan x}{1 - \tan x} \neq -\left(\dfrac{\tan x}{1 + \tan x}\right)$

$\neq \left(\dfrac{\tan x}{1 + \tan x}\right)$

neither odd nor even #

Example 1.5.10 *Determine the domain and range of the function:*

$$f(x) = \begin{cases} 1 - x & , x < 1 \\ \sqrt{x - 1} & , x \geq 1 \end{cases}$$

Sol

Exercise 1.5.1 *1. Evaluate (if possible) the function at the given values of the independent variable. Simplify the result*

(a) $f(x) = 5 - x^2$

(i) $f(0)$

(ii) $f(-3)$

(iii) $f(b)$

(iv) $f(x - 1)$

(b) $f(x) = \cos 2x$

(i) $f(0)$

(ii) $f\left(-\frac{\pi}{4}\right)$

(iii) $f\left(\frac{\pi}{3}\right)$

2. *Find the domain and range of the functions:*

   (i) $g(x) = x^2 - 5$

   (ii) $f(x) = 4x^2$

   (iii) $h(x) = \dfrac{3}{x}$

   (iv) $g(x) = \dfrac{2}{x-1}$

3. *Find the domain of definition of the following functions:*

   (i) $f(x) = \sqrt{x} + \sqrt{1-x}$

   (ii) $g(x) = \dfrac{1}{\sin x - \frac{1}{2}}$

   (iii) $f(x) = \dfrac{1}{|x^2 - 4|}$

   (iv) $f(x) = 2x^3 + \dfrac{4}{x} + 5$

   (v) $g(x) = x^7 + \dfrac{3}{\sqrt{x}} - x^2$

   (vi) $h(x) = \dfrac{2+x}{\sqrt{x} - 3}$

   (vii) $f(x) = \left(\dfrac{x-3}{x-8}\right)^{\frac{1}{2}}$

   (viii) $f(x) = x^2 + 1 + \sqrt{x-2}$

   (ix) $f(x) = x^2 + x + \dfrac{2}{x1}$

   (x) $g(x) = x^2 + x^{-3} + 3$

4. *Given $f(x) = \sin x$ and $g(x) = \pi x$, evaluate each expression*

   (i) $f\left(g(2)\right)$

   (ii) $f\left(g\left(\dfrac{1}{2}\right)\right)$

   (iii) $g\left(f(0)\right)$

   (iv) $g\left(f\left(\frac{\pi}{4}\right)\right)$

   (v) $f\left(g(x)\right)$

   (vi) $g\left(f(x)\right)$